

Clinical Risk Assessment Report: Patient 12

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 34.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.57x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +5.2%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 1.31 [Top 8%] (+5.2%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.77 [Top 14%] (+3.2%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.63 [Avg (65th %ile)] (+3.0%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 2.31 [Top <1%] (+2.4%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.27 [Avg (33th %ile)] (+2.0%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: -1.05 [Bottom 15%] (+1.4%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.18 [Avg (32th %ile)] (+1.0%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 74.0 [Top 13%] (-2.5%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Machine Learning Feature Attribution (SHAP Decision Path)

